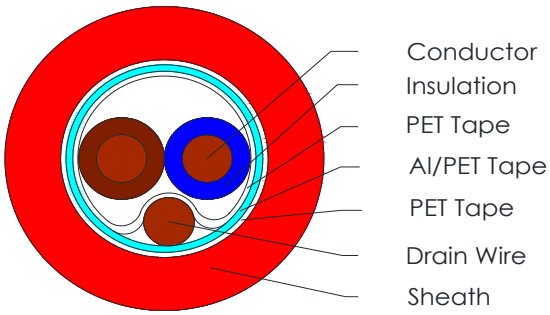
Fire Resistant Cables - Fireplus Two core, Overall Screen, LSZH/HFFR Sheath BS 6387 CWZ, EN 50200 PH120, UL Certified

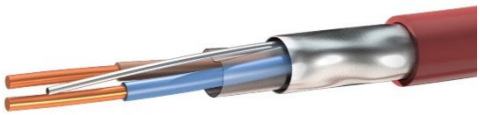
C91044, C91045, C91283, C91284, C91289, C95126

Applications

Screened two core Fire Resistant cable for Building and Industrial Management Systems. UL Certified E539056.

Cross Section Drawing





Design

Unit	Properties
Conductor	2 x Solid or Flexible Bare Copper wire
Insulation	Ceramifiable Silicon Rubber Core 1: Blue Core 2: Brown
Wrapping tape	PET tape
Drain Wire	Solid Tinned Copper wire
Screen	Aluminium/Polyester tape
Wrapping tape	PET tape
Sheath Material	Halogen Free Flame-Retardant (LSZH/HFFR) Standard Colour: Red (-02) or White (-09)
Standard Put Up Length	305 and 500 metres

*Other Colors, Put Up Lengths and structures can be manufactured upon request, please contact your local B3 International sales representative.

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Physical Characteristics

Part Number	C91284	C91283	C91044*	C91045*	C91289*	C95126*
No of cores x cross section in sqmm (mm2)	2 x 0.75	2 x 1.0	2 x 1.5	2 x 2.5	2 x 4.0	2 x 6.0
Nom. Diameter Conductor (mm)	1 x 0.98	1 x 1.13	1 x 1.38	1 x 1.78	7 x 0.83	84 x 0.285
Nom. Radial Thickness Insulation (mm)	0.7	0.6	0.7	0.8	0.8	1.0
Twist per meter	12	11	10	8	6	5
Nom. Cross Section Drain Wire (mm2)	0.50	0.50	0.50	0.50	0.50	0.50
Drain Wire Stranding	1x0.8	1x0.8	1x0.8	1x0.8	1x0.8	1x0.8
Screen Coverage (%)	115					
Nom. Overall Diameter (mm)	7.0	7.3	8.1	9,30	11.8	12.8
Cable weight (kg/km)	67	75	90	120	201	254
Operating Temperature (°C)	-20 to +90 (3hrs 950°C)					
Installation Temperature (°C)	-15 to +90					
Minimum bending radius (mm)	75	78	80	99	120	140
Max. recommended pulling tension (N)	205	265	405	670	1250	1875
Fire Resistance to BS6387, Cat. C	Exposed to fire at 950°C for 3 hours					
Fire Resistance to BS6387, Cat. W	Exposed to fire at 650°C for 15 minutes, then exposed to fire at 650°C with water for 15 minutes					
Fire Resistance to BS6387, Cat. Z	Exposed to fire at 950°C for 15 minutes, then exposed to fire at 650°C with mechanical shock for 15 minutes					
Fire Resistance to IEC 60331-21	Exposed to fire at 750°C for 90 minutes					
Flame Retardancy	IEC 60332-3-24					

Electrical Characteristics at 20°C

Part Number	C91284	C91283	C91044*	C91045*	C91289*	C95126*
Max. DC Resistance Conductor (Ω/km)	24.5	18.1	12.1	7.41	4.61	3.08
Nom. Capacitance conductor to conductor (pF/m)	87	95	105	125	125	147
Nom. Capacitance conductor to rest (pF/m)	160	170	185	200	195	220
Min. Insulation Resistance (MΩ*km)	200					
Max. recommended current at 25°c (Amps)	12	18	21	30	40	50
Max. Operating Voltage (Vrms)	300/500					

Reference Standards

BS EN 50363-1	*EN 50200 PH120
EN 50290-2-27	*IEC 60754-1
IEC 60228/BS6360	IEC 60754-2
IEC 60332-3-24	*IEC 61034-2
IEC 61034-1	BS 7671
IEC 60331-21 FE180	BS 7655.6-1
*BS 6387 CWZ	RoHS Directives

*Structure&Standards verified under UL Certification E539056